Introduction to Statistical Methods

(S1-26_AIMLCZC418) – Assignment 1

Section: **Prof. G Krishna Kumari**

Each question carries 2.5 Marks (2.5 x 4 = 10 Marks)
Duration: 06th September 2026 – 30th September 2026
1) Submissions are individual
2) Solve these on paper, scan, and upload
3) Plagiarism results in zero marks
4) Write your name, BITS ID and Section on each page
5) Only handwritten solutions with formula, full steps with proper justification are required.

Q1. The completion times (in days) for 12 recent software modules are:
14, 16, 15, 18, 17, 16, 15, 14, 17, 16, 15, 72

a) Compute the mean, median, sample variance, range, and Q1, Q3 for this group.
b) Determine the skewness (left/right) based on central tendencies.
c) Recommend which measure the project manager should report to clients and justify why.

Q2. Among 48 research scholars, 20 collaborated with other institutes, 18 participated in international conferences, and 8 did both. If a scholar is picked at random, what is the probability they collaborated with other institutes or participated in a conference?

Q3. You have the following dataset:

Patient	Fatigue	Loss of Smell	Sore Eyes	Has Flu
P1	Yes	Yes	No	Yes
P2	No	No	Yes	No
P3	Yes	No	No	Yes
P4	No	Yes	Yes	No

A new patient exhibits: Fatigue = No, Loss of Smell = Yes, Sore Eyes = No.

Apply Naive Bayes to predict if the patient has flu.

Q4. A factory uses an IoT monitoring system to detect abnormal sensor readings in its machines.

Actual anomalies are rare — only 1% of all readings come from truly abnormal machine behaviour.

The anomaly-detection system works as follows:

- When a real anomaly occurs, the system correctly flags it 88% of the time.
- When everything is normal, the system still raises a false alarm 6% of the time.

One of the latest sensor readings has been flagged as anomalous by the system.

What is the probability that the reading is actually from a real anomaly?

----ALL THE BEST----